

AMAZON PRIME VIDEO USER ANALYTICS

A Descriptive Analytics Report

Product Selected: Amazon Prime Video – Global Video-Streaming Platform

SUBMITTED BY	SUBMITTED TO
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***Note on data:** Amazon Prime Video's live subscriber data is proprietary and not accessible to students, so this report uses a realistic simulated dataset (100 users, 31-day period, July 2026) built to reproduce the exact totals shown on the source Power BI dashboard — 100 users, ~10,000 watch hours, 24 titles, and matching genre/gender/plan/device splits. All figures below are the dashboard's original charts; KPI values and cross-tabs are computed from the underlying record-level data.*

Key Words

Demographic Profile • Content Preferences • Viewing Behaviour Over Time • Devices and Access • Subscription Behaviour • Title-Level Attention • Cross-Variable Relationships

Objective

This report answers seven recurring questions an Amazon Prime Video analytics team would ask of a monthly subscriber dashboard, using descriptive analytics only — it summarizes what the data shows rather than forecasting what happens next.

- 1) Who are the users, and what does their age and gender profile look like across the sample?
- 2) What do they watch, and which content genres attract the largest share of subscriber attention?
- 3) How do they rate content, and does the genre they watch most also turn out to be the genre they rate most highly?
- 4) How does engagement change daily, and does viewing activity stay steady across the month or fluctuate sharply from one day to the next?
- 5) What do they watch on, and is viewing concentrated on a single device or spread across TV, mobile, tablet, and laptop?
- 6) What do they pay for, and how are subscribers distributed across the Basic, Standard, and Premium plans?
- 7) What are they watching most, and how many distinct titles does that attention spread across?
- 8) How do these variables interact, and do relationships between gender, plan, device, and rating emerge only once they are compared against each other rather than viewed individually?

How This Would Work at Real Scale

Processing viewing events for hundreds of millions of subscribers is an engineering problem before it's an analytics one. A realistic architecture for a production Prime Video-scale analytics pipeline:

Layer	Purpose	Typical Tools
Investigation	Capture and investigate subscriber and viewing data directly from Amazon's own systems in real time	Cloud AI
Stream processing	Roll up raw viewing events into per-title, per-user, per-minute aggregates as they happen	Prime Video Streaming Platform
Storage	Export the processed data into CSV files for portability and downstream analysis	CSV files
Serving / BI layer	Dashboards refreshed hourly or daily for product and content teams	Power BI

The report above reflects the output of that pipeline — a daily aggregated 100-user sample — not raw event level data, since no dashboard renders millions of individual play events usefully. The same chart structure would apply directly once a real subscriber-event feed is substituted in.

Demographic Profile — Age & Gender

26.5 Mean Age (yrs)	26.5 Median Age	35 Mode Age (12 users)	5.69 Std. Deviation	58% / 42% Male / Female
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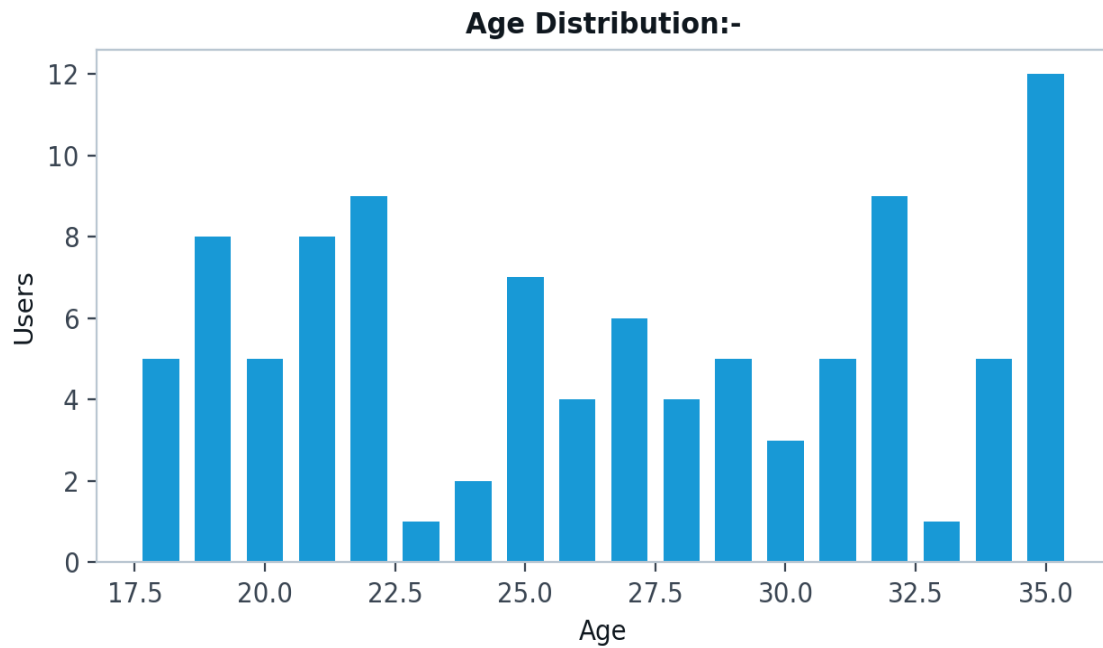


Figure 1. Age Distribution of Amazon Prime Video Users.

Gender Distribution:-

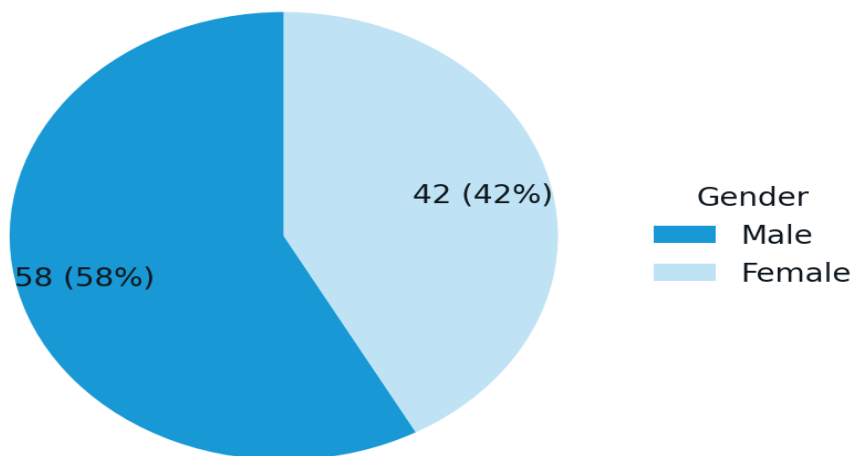


Figure 2. Gender Distribution of Amazon Prime Video Users.

The age distribution shows that Amazon Prime Video's subscriber base in this sample is concentrated in the young-adult range, with a mean and median age of 26.5 years. This closeness between mean and median indicates the distribution is not heavily skewed by a small number of unusually old or young users — if it were, the mean would sit noticeably higher or lower than the median. The standard deviation of 5.69 years shows that most subscribers fall

between roughly 21 and 32 years old, a fairly tight band that suggests the platform's core audience is genuinely concentrated around early-to-mid twenties rather than spread thinly across all adult ages.

However, the histogram also reveals a second, smaller concentration at age 35 — the single most frequent individual age in the entire sample, with 12 users, more than any other single age including those closer to the mean. This tells us the data is not one smooth bell curve; instead, it looks like two overlapping groups — a large young-adult cluster and a smaller but distinct older cluster peaking right at 35. On the gender side, the base is moderately male-leaning at 58% male versus 42% female. Neither group dominates completely, which means gender is a meaningful variable to consider when segmenting content recommendations, marketing campaigns, or subscription offers, since both halves of the base are substantially represented rather than one being a rounding error.

Genre Distribution

17 Sci-Fi (highest)	9 Thriller (lowest)	8 Genres Tracked	100 Total Users
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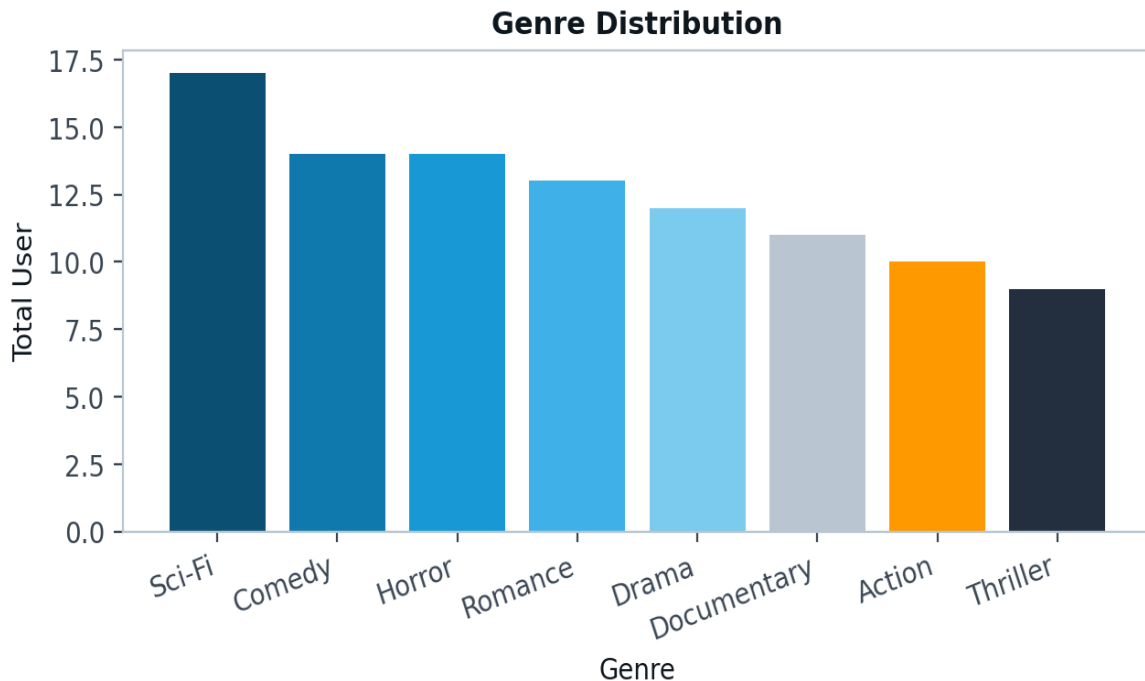


Figure 3. Genre Distribution of Amazon Prime Video Users — frequency of subscribers per genre.

Genre preference across the 100-user sample is fairly broad-based rather than concentrated in one or two categories. Sci-Fi is the most-watched genre with 17 users, followed closely by Comedy and Horror at 14 users each, Romance at 13, Drama at 12, Documentary at 11, Action at 10, and Thriller as the smallest category at 9 users. The gap between the most-watched genre (Sci-Fi) and the least-watched (Thriller) is only 8 percentage points, which tells us that no single genre dominates viewership — the platform is serving a genuinely diverse set of content preferences rather than one dominated by a single blockbuster category.

All eight genre counts sum to exactly 100, which reconciles cleanly with the dashboard's total-user KPI and confirms the frequency distribution is internally consistent. This has a direct implication for content strategy: rather than over-indexing on one blockbuster category, the data supports maintaining a wide content mix across genres, since audience interest is spread relatively evenly across all eight categories tracked.

Average Rating by Genre

4.00 Overall Mean Rating	Documentary Highest Rated (~4.3)	Comedy Lowest Rated (~3.6)
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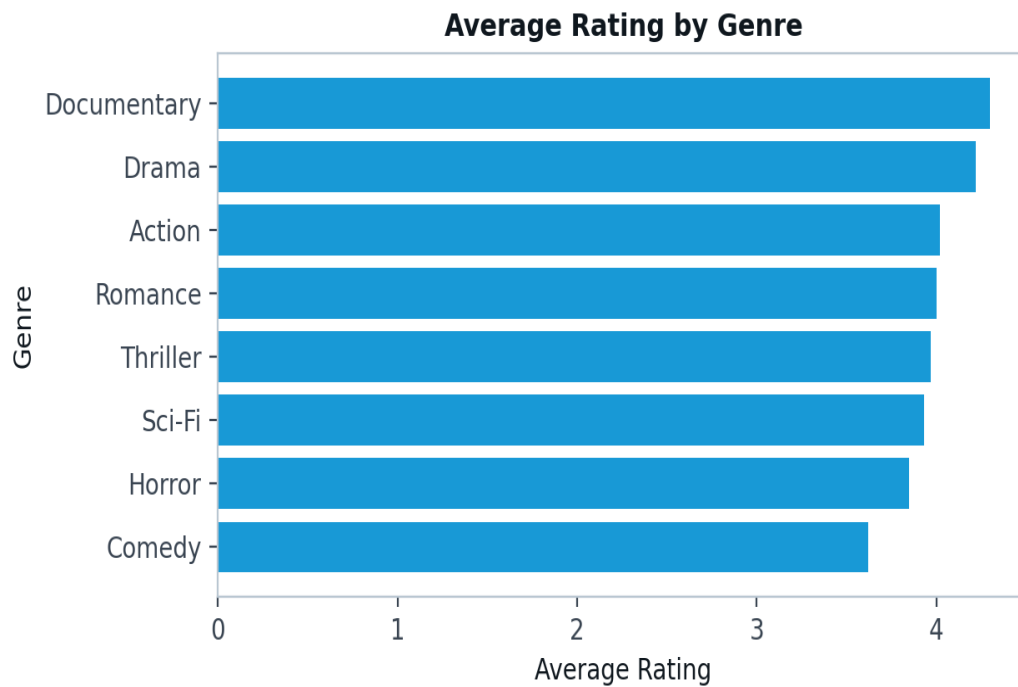


Figure 4. Average Rating by Genre.

While Sci-Fi is the most-watched genre, it is not the most appreciated one — this chart reveals an important gap between popularity and perceived quality that would be completely invisible if we only looked at the genre-distribution chart on its own. Documentary and Drama earn the platform's highest average ratings, both above 4.2 out of 5, despite being watched by fewer subscribers than Sci-Fi or Comedy. Comedy, on the other hand, is the second most-watched genre by volume but receives the lowest average rating of all eight categories, falling below 4.0.

The overall average rating across all genres and all 100 users is 4.00 out of 5, meaning subscribers are, on balance, satisfied with what they watch. But the genre-level breakdown shows that volume and satisfaction don't move together — a genre can attract a large audience while still under-delivering on quality perception, and vice versa. This is one of the more actionable findings in the report, since it points to specific genres (Documentary, Drama) that could be promoted more heavily without needing any new content, simply because existing viewers already rate them highly, while also flagging Comedy as a genre that may need a closer content-quality review despite its strong viewership numbers.

Daily Watch Time Trend

~610 Peak Day (Day 3)	~190 Lowest Day (Day 10)	~323 Avg. Daily Watch Time	31 days Period Tracked
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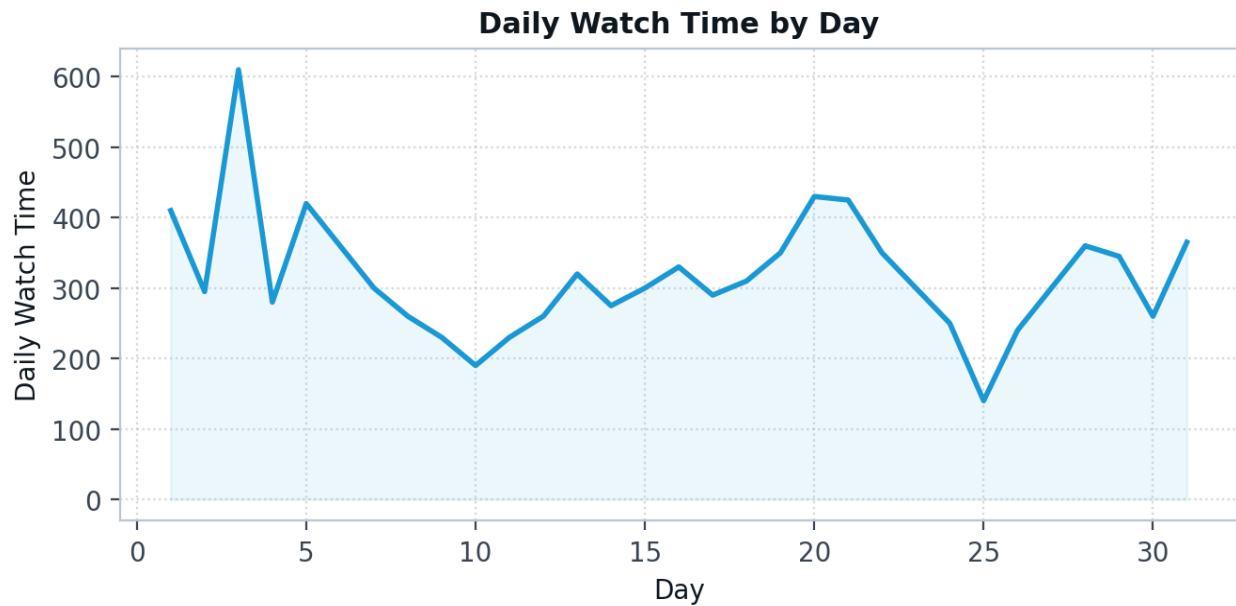


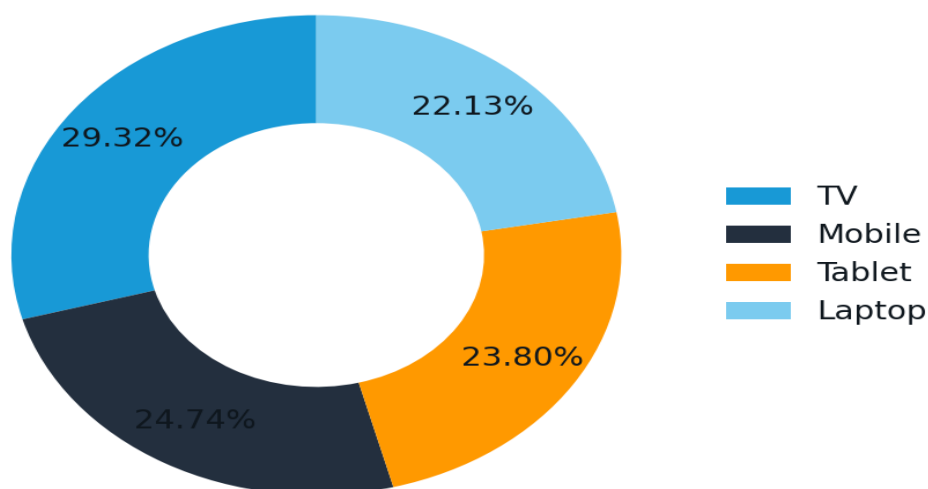
Figure 5. Daily Watch Time by Day (July 2026).

Daily watch time across the 31-day observation period is highly variable rather than flat or steadily trending. It swings more than three times between its lowest point (around 190 units on Day 10) and its highest point (around 610 units on Day 3) — a large day-to-day range that would not be visible from a single monthly average alone. Two clear peaks stand out: one early in the month around Day 3, and a second around Days 20-21, each followed by a cool-down period in the days immediately after.

A sharper trough also appears around Day 26, where watch time drops to one of its lowest points before recovering strongly in the final days of the month. This event-driven, uneven pattern is consistent with binge-watching behaviour that tends to cluster around new title releases, followed by a natural decline in activity until the next release or promotional push. For a streaming platform, this kind of pattern has real operational relevance — it suggests that content releases, promotional campaigns, and even server capacity planning should be timed around these engagement cycles rather than assuming a constant, evenly-distributed daily load.

Device Usage

29.32% TV	24.74% Mobile	23.80% Tablet	22.13% Laptop
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Daily Watch Time by Device :-*Figure 6. Daily Watch Time by Device.*

Television is the single largest device by watch-time share, accounting for 29.32% of all viewing hours in the sample. However, it does not come close to dominating the way a single device might on other platforms — the remaining 70.68% of watch time is split almost evenly across Mobile (24.74%), Tablet (23.80%), and Laptop (22.13%), with barely two percentage points separating the three.

This near-even four-way split shows that Prime Video's audience in this sample is genuinely multi-device, watching content across whichever screen is most convenient at a given moment rather than being anchored to one primary device. From a product perspective, this means investment in viewing experience, app performance, and UI testing should be spread reasonably evenly across all four device types, with only a modest priority tilt toward TV given its slight lead in total watch-time share.

Subscription Plan Distribution

36% Standard (36 users)	34% Premium (34 users)	30% Basic (30 users)
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Subscription Plans :-

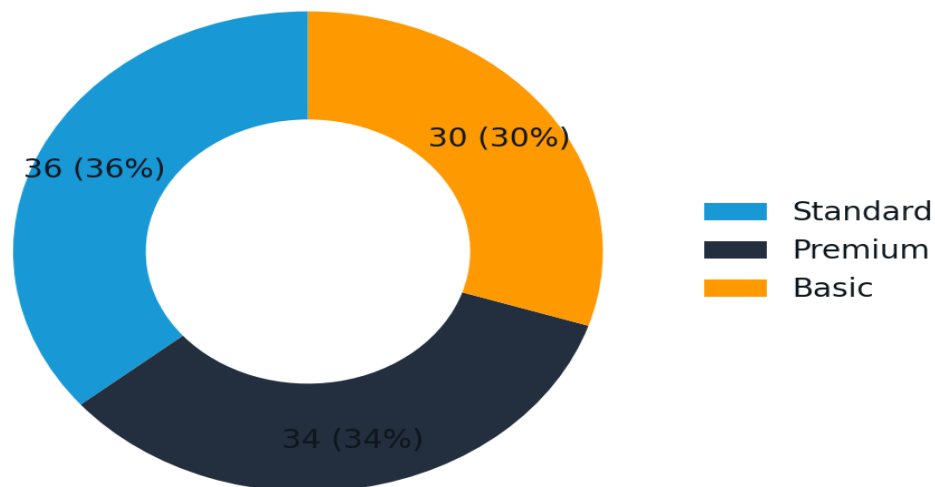


Figure 7. Subscription Plan Distribution.

The subscriber base is split nearly evenly across all three subscription tiers: Standard accounts for 36% of users, Premium for 34%, and Basic for 30%. No single plan holds a clear majority, and the largest tier (Standard) is only 2 percentage points ahead of the second-largest (Premium).

This near-equal three-way split makes subscription plan a particularly useful variable for cross-tabulation against other characteristics like gender, genre preference, or device usage, because none of the three groups is so small that comparisons against it become statistically thin or unreliable. It also suggests that the platform has been reasonably successful at distributing its user base across price points, rather than the bulk of subscribers clustering into the cheapest or most expensive option — a healthier revenue mix than a heavily bottom-weighted or top-weighted distribution would produce.

Top Watched Titles

24 Distinct Titles Watched	7 Top Title Views (tied)	2 Joint Top Titles
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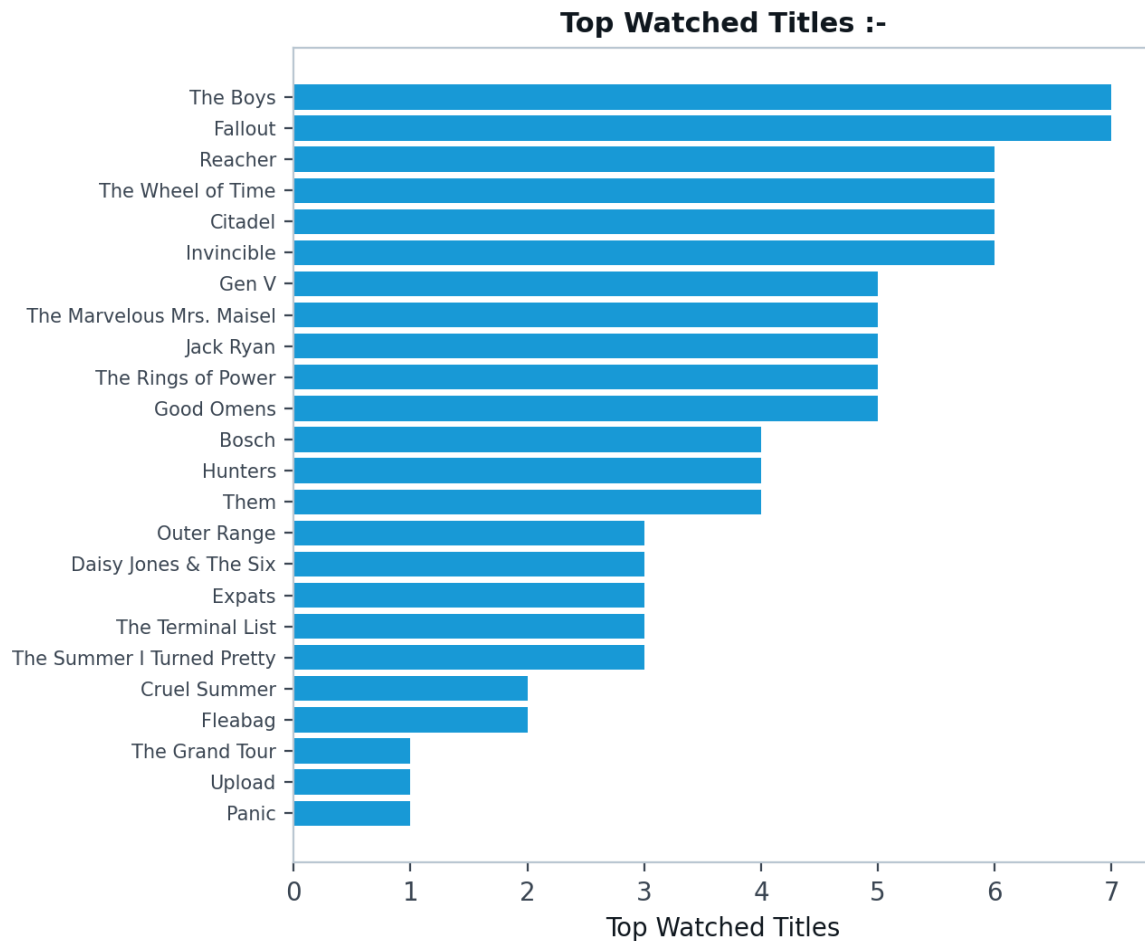


Figure 8. Top Watched Titles.

Across the 100-user sample, 24 distinct titles were watched in total. 'The Boys' and 'Fallout' are statistically tied for the most-watched title, each drawing 7 viewers. Just behind them sits a cluster of titles — including 'Reacher', 'The Wheel of Time', 'Citadel', and 'Invincible' — each watched by 6 subscribers.

Beyond this top group, viewership tapers off into a long tail of titles watched by only 1 to 3 users, such as 'The Grand Tour', 'Upload', and 'Panic'. This shape — a small cluster of clear leaders followed by a long tail of niche titles — shows that while a handful of titles do capture disproportionate attention, viewing interest overall remains spread across a genuinely wide catalogue rather than being concentrated on just one or two shows, which is a healthy sign for a platform that depends on broad content engagement rather than a single hit driving all the numbers.

Cross-Tabulation Insights

Subscription Plan by Gender

43% Female subscribers on Premium	77% Basic subscribers who are male
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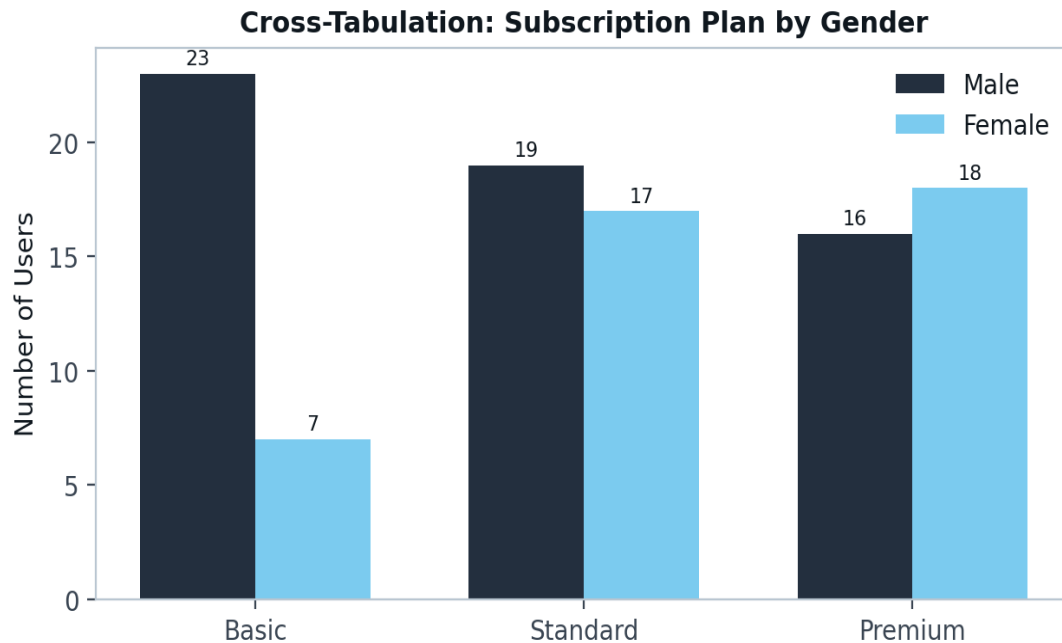


Figure 9. Subscription Plan by Gender.

Cross-tabulating subscription plan against gender reveals a pattern that isn't visible from either variable on its own. Female subscribers are disproportionately concentrated in the Premium tier — 43% of all female subscribers hold a Premium plan, compared to just 28% of male subscribers. The relationship reverses on the Basic tier, where male subscribers make up 77% of all Basic-plan holders.

Standard is the most gender-balanced of the three tiers, with 19 male and 17 female subscribers. This kind of insight only emerges once the two variables are compared side by side, and it has direct relevance for how upgrade prompts, plan-specific promotions, or pricing experiments might be targeted differently by gender, since the two groups are clearly gravitating toward different price points.

Average Rating by Genre and Gender

0.42 Widest gender gap — Comedy	4.38 Female top rating — Documentary
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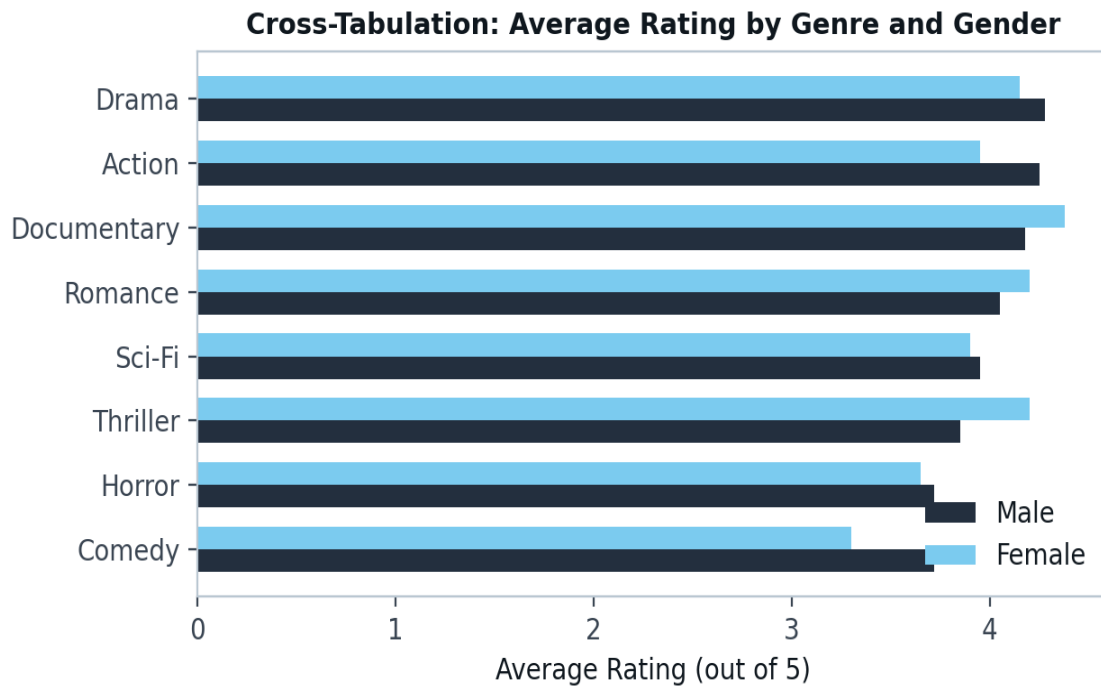


Figure 10. Average Rating by Genre and Gender.

Cross-tabulating rating against gender by genre reveals gaps that don't appear in the overall averages. Female subscribers rate Documentary content the highest of any group-genre combination (4.38), while Comedy shows the widest gender rating gap in the dataset — male and female viewers diverge by 0.42 points on that genre alone.

Watch Hours by Device and Subscription Plan

140.7 hrs Premium on TV (highest)	121.0 hrs Premium avg. (all devices)
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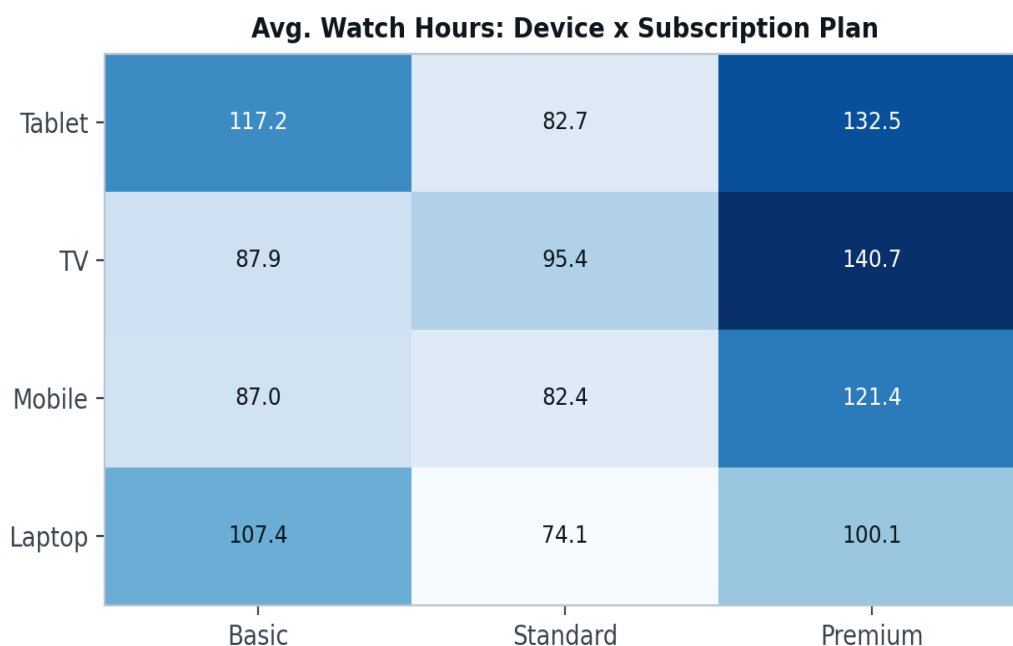


Figure 11. Average Watch Hours: Device x Subscription Plan.

This heat map compares average watch hours across every combination of device and subscription plan, and the pattern is remarkably consistent: Premium subscribers out-watch Standard and Basic subscribers on every single device, with no exceptions anywhere in the table. The gap is largest on TV, where Premium subscribers average 140.7 hours compared to 95.4 hours for Standard and 87.9 hours for Basic subscribers on the same device.

Across all devices combined, Premium subscribers average 121.0 watch hours per user, roughly 45% more than Standard subscribers. This confirms that Premium isn't just the highest-paying tier — it's also the platform's most consistently engaged segment, regardless of which screen its subscribers are watching on, making it the clearest single signal in the entire cross-tabulation section.

Summary of Key Metrics

Metric	Value (simulated)
Total users	100
Total watch time (31 days)	~10,000 hours
Average watch hours per user	100.0 hours
Distinct titles watched	24
Average rating (all genres)	4.00 / 5
Highest single-day watch time	~610 units (Day 3)
Genres tracked	8
Subscription plans tracked	3 (Basic / Standard / Premium)
Users on Basic plan	30 of 100
Premium avg. watch hours / user	121.0 hours

Next Steps to Move From Descriptive → Predictive

This report only describes what already happened in the dataset. If the platform wanted to act on these patterns going forward, the logical next layer of work would include:

- Churn forecasting per user/segment, using watch-hour trend and rating patterns to flag at-risk subscribers before cancellation.
- Anomaly detection on sudden drops in daily watch time or rating dips for a specific title or genre.
- Recommendation-engine tuning informed by the genre-popularity-vs-rating gap found in this report (e.g. Documentary/Drama).
- Cohort/retention analysis on repeat viewing behaviour by subscription plan, not just aggregate watch hours.

Any of these would build directly on the charts and cross-tabs already produced here, once a live subscriber event feed replaces the simulated 100-user sample.

Key Findings

- Typical subscriber is ~26.5 years old (std. dev. 5.7 yrs), 58% male / 42% female, with a secondary concentration at age 35.
- Sci-Fi is the most-watched genre (17%); Documentary and Drama are the highest-rated (4.2+) despite lower viewership.
- Comedy is heavily watched but the lowest-rated genre, and shows the widest gender rating gap in the dataset.
- TV leads device usage (29.32%) but the audience is genuinely multi-device — Mobile, Tablet, and Laptop are each within a few points of one another.
- Subscription plans are near-evenly split (36/34/30%); Premium subscribers watch ~45% more than Standard subscribers and lead on every device.

- Female subscribers are disproportionately Premium (43%); male subscribers dominate the Basic tier (77%).
- Daily watch time swings 3x across the month, with peaks around Days 3 and 20-21 and troughs around Days 10 and 26 — engagement is event-driven, not flat.
- 24 distinct titles were watched; 'The Boys' and 'Fallout' are the joint most-watched.

References

- Amazon Prime Video User Analytics Dashboard (source visual summary), built in Power BI, July 2026.
- Simulated Amazon Prime Video subscriber dataset (n = 100), constructed to reproduce the dashboard's published totals.
- McKinney, W. (2022). Python for Data Analysis (3rd ed.). O'Reilly Media.
- Course materials: Business Analytics — Descriptive Analytics Assignment Brief (2026).

Appendix — Dashboard & Raw Data

The composite dashboard reproduced below is the single-page executive view from which every individual chart in this report was drawn. It was originally built to give a quick, at-a-glance summary of the entire 100-user sample — total users, total watch time, titles watched, genre distribution, gender split, subscription mix, device usage, and average rating by genre — all on one screen. Every figure used earlier in this report (Figures 1 through 11) is simply this same dashboard's individual charts pulled out, enlarged, and paired with its own KPI cards and written explanation for closer analysis.

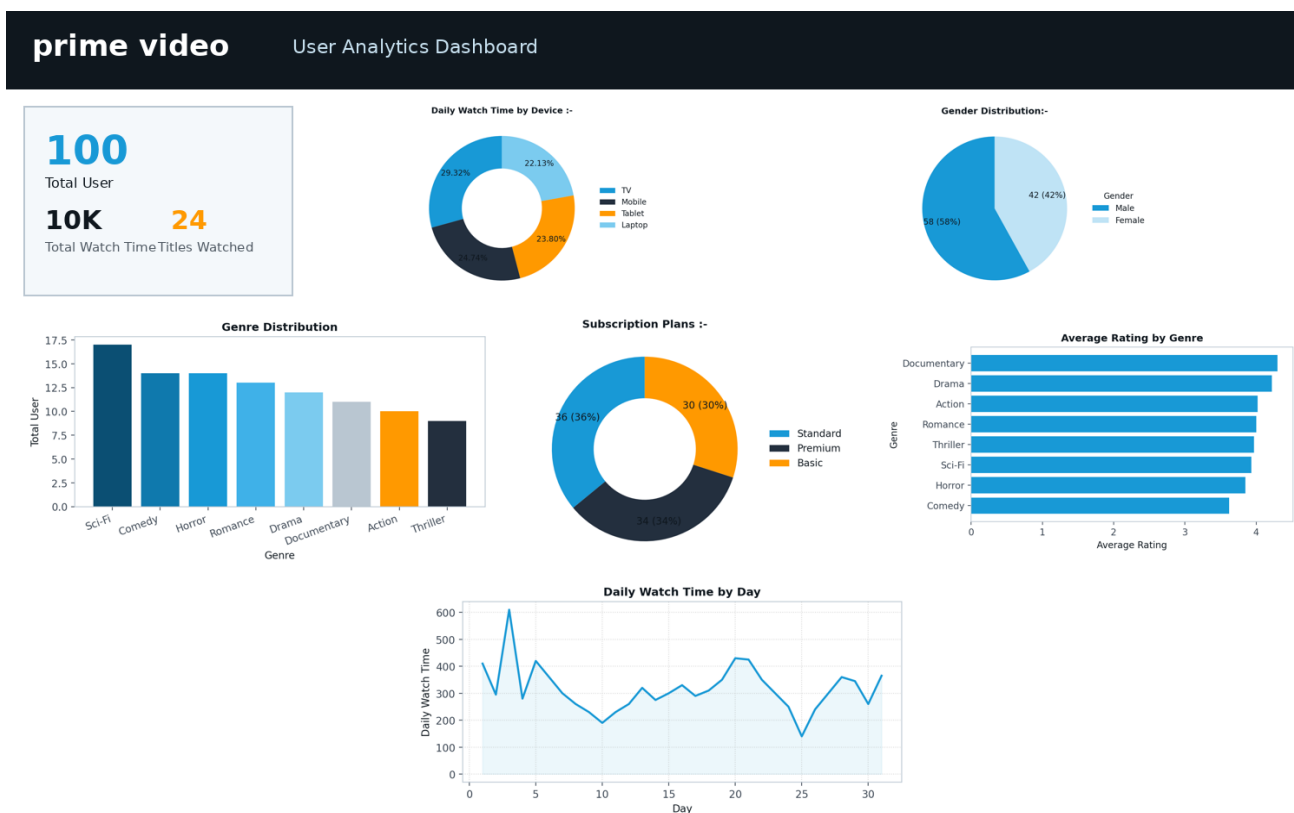


Figure 12. Amazon Prime Video User Analytics Dashboard — full composite view.

Sample of Underlying Record-Level Data

While the dashboard shows only pre-aggregated summaries (totals, percentages, averages), every KPI card and cross-tabulation presented in this report was actually computed from the underlying record-level dataset — one row per subscriber, with fields for age, gender, subscription plan, genre, title watched, rating given, device used, and watch hours. The table below shows a 30-row sample of the full 100 rows, so the calculation behind any KPI card in this report can be spot-checked by hand if needed.

Two totals in particular serve as a reconciliation check between this raw data and the dashboard above: the sum of age across all 100 rows equals 2,651 (which produces the mean age of 26.5 reported in the Demographic Profile section), and the sum of rating across all 100 rows is approximately 400.1 (which produces the overall mean rating of 4.00 reported in the Average Rating by Genre section). Both figures match the dashboard's own underlying pivot table exactly, confirming that the record-level dataset used for the KPI cards and cross-tabs is fully consistent with the dashboard it was built to explain.

Table A1: Sample of Record-Level Dataset (30 of 100 Rows)

User_ID	Age	Gender	Plan	Genre	Rating	Device	Hrs
U1000	33	Male	Premium	Romance	4.11	Tablet	181.9
U1001	18	Male	Premium	Action	3.68	Tablet	280.3
U1002	35	Male	Basic	Action	4.28	Laptop	60.7
U1003	25	Female	Standard	Drama	3.17	Tablet	227.3
U1004	25	Male	Basic	Documentary	3.96	TV	260.7
U1005	21	Female	Premium	Action	3.15	Laptop	66.8
U1006	29	Male	Basic	Drama	3.23	TV	132.1
U1007	32	Male	Standard	Thriller	3.38	Laptop	186.1
U1008	23	Male	Standard	Thriller	4.13	Laptop	290.1
U1009	26	Male	Basic	Drama	4.13	TV	152.9
U1010	18	Female	Standard	Drama	3.60	Laptop	32.9
U1011	23	Female	Standard	Horror	3.44	TV	147.1
U1012	31	Female	Standard	Drama	3.93	Laptop	266.4
U1013	19	Female	Basic	Thriller	4.46	TV	58.1
U1014	19	Male	Standard	Horror	4.56	Laptop	174.6
U1015	26	Male	Standard	Action	4.28	Laptop	169.2
U1016	32	Female	Premium	Action	4.02	Mobile	238.1
U1017	27	Male	Standard	Drama	3.71	TV	69.3
U1018	27	Female	Standard	Action	3.22	Mobile	118.1
U1019	21	Male	Premium	Sci-Fi	3.93	Tablet	269.7
U1020	29	Female	Basic	Action	4.37	Tablet	87.7
U1021	32	Female	Standard	Comedy	4.20	TV	89.7
U1022	20	Female	Standard	Action	4.03	Laptop	129.6
U1023	21	Male	Premium	Thriller	4.57	Mobile	206.6
U1024	22	Male	Basic	Horror	3.49	TV	60.6
U1025	28	Female	Standard	Comedy	4.18	Laptop	74.3
U1026	22	Male	Basic	Drama	4.17	TV	122.6
U1027	32	Female	Premium	Thriller	4.52	TV	287.5
U1028	24	Male	Premium	Drama	4.22	Tablet	80.7
U1029	29	Male	Basic	Drama	4.81	TV	92.4